7.1 Introduction

Instrumentation is the heart of an industry for the measurement of the parameters like pressure, temperature, force, displacement and rotation etc. A basic instrumentation system consists of major devices for its operation. One of these categories is transducers and it acts as major role in the measurement and instrumentation systems.

Transducers are energy conversion devices which transforms non-electrical quantity into electrical quantity. The transducers are very important in our day to life to measure the electrical parameters such as current, voltage etc., and physical parameters are pressure, load, torque etc.

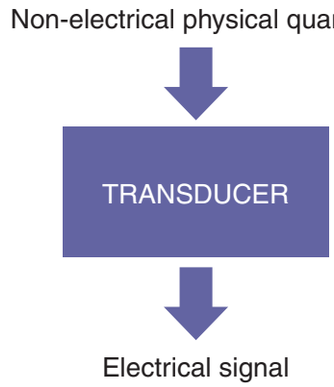


Fig. 7.1 Transducer

7.2 Principle of operation of transducers

All electrical instrumentation system consists of operating elements to perform a measurement and to display its results. The major elements of an instrumentation system is

i. Input devices
ii. Processing devices and
iii. Output devices

The input devices receive the quantity under measurement. The processing devices transmit the signal from the input devices to the output devices. The output device gives the value of output sensed by the transducers. So, in a broad category the transducers convert the non-electrical quantity such as temperature, pressure, sound etc into electrical quantity like voltage, current and capacitance etc.

7.3 Classification of transducers

The transducer can be classified as,

i. On the basis of transduction form used
ii. Primary and secondary transducers
iii. Passive and active transducers
iv. Analog and digital transducers
v. Transducers and inverse transducers

i. The transducers are classified based on the method of transduction such as resistive, inductive and capacitive

For resistive transducers, variation of input causes the resistance to change.

For inductive transducers, variation of input causes the inductance to change.

For capacitive transducers, variation of input causes the capacitance to change.

ii. Primary and Secondary transducers

The transducer connected on the detector stage is called Primary transducer and the output of primary transducer is

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